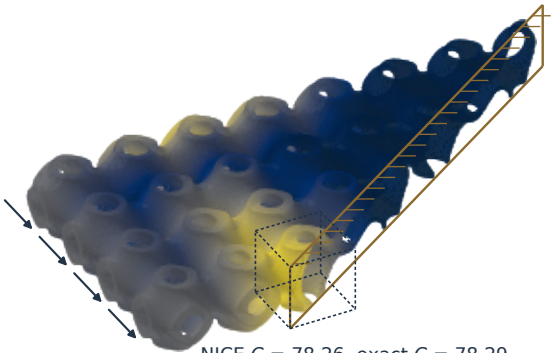
(a) Final NICE design (iteration 23)

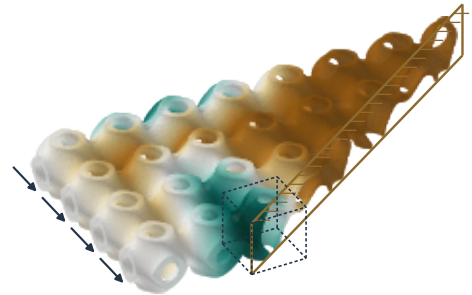


NICE C = 78.26, exact C = 78.29



local thickness parameter $\tau(x)$ on the walls

(b) Change from the uniform start

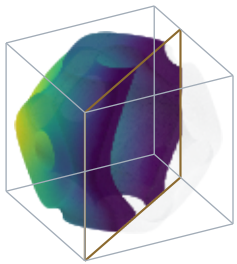




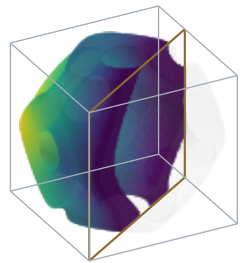
change of thickness parameter, $\tau - 0.40$

(c) Cut cell next to the support (Figure 1b) at the final design, design load: displacement magnitude on the walls

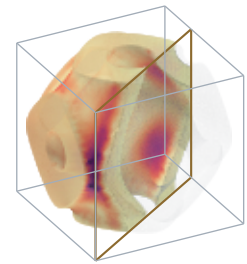
NICE, $|\widehat{\Psi}_m B_m \widehat{U}|$



exact condensation, $|\Psi_m B_m U|$



difference $|\widehat{\Psi}_m B_m \widehat{U} - \Psi_m B_m U|$





displacement magnitude $|u|$ (shared scale)



$|u_{\text{NICE}} - u_{\text{exact}}|$ (own scale)

Relative difference of the cell field, NICE vs exact: 1.04% in the energy norm, 0.76% in the nodal ℓ_2 norm;
largest difference on the walls 0.29% of the largest $|u|$ on the walls